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Posted Date: 23 September 2026

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Article

Constraint-Guided LLM-GNN Framework for Anomalous Journal Entry Detection in Auditing

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Abstract

Journal Entry Testing (JET) is a fundamental audit procedure to identify any potential misstatements, fraud, and management override of controls. Traditional rule-based JET methods suffer from high false positive rates and limited ability to detect complex anomaly patterns. Recent work has shown that large language models (LLMs) can serve as effective anomaly detectors for bookkeeping data, but LLM-only approaches lack explicit modeling of accounting constraints and structural relationships among entries. We propose the Constraint-Guided LLM-GNN (CG-LGN) framework, which integrates three complementary modules: (1) a heterogeneous graph neural network (GNN) that models structural relationships among journal entries, accounts, users, and temporal attributes; (2) an accounting constraint module encoding domain-specific rules such as debit-credit integrity, unusual account combinations, and period-end concentration; and (3) an LLM-based explanation generator that produces auditor-readable interpretations for flagged entries. Experiments on synthetic journal entry data with six injected anomaly types show that CG-LGN achieves a PR-AUC of 0.49 (a 0.20 absolute gain over the strongest single-module baseline) and reduces false positives per 1,000 entries by 47%. Ablation studies confirm that the GNN and constraint modules improve detection performance, while the LLM module improves explanation quality.

Keywords: journal entry testing; anomaly detection; graph neural network; large language model; accounting constraints; financial auditing

I. Introduction

Journal entries constitute the atomic records of financial accounting systems. Every business transaction is recorded as a journal entry that specifies one or more debit and credit lines across accounts, subject to the fundamental constraint that total debits equal total credits. Journal Entry Testing (JET) is the auditing procedure through which auditors examine the full population of journal entries to identify suspicious patterns that may indicate errors, fraud, or intentional manipulation [1].

Traditional JET relies on predefined rules such as flagging entries posted after business hours, large round-number amounts near approval thresholds, or unusual account combinations [2]. While effective for known fraud patterns, rule-based JET produces a high volume of false positives. Machine learning approaches, including Isolation Forest [3] and deep autoencoders, have improved detection accuracy by learning distributional patterns, yet they remain limited in capturing the rich relational structure inherent in accounting systems.

More recently, Kadir et al. [4] benchmarked LLMs against traditional JET rules and ML baselines for fraud detection in double-entry bookkeeping, demonstrating that LLMs can match or surpass classical methods while providing natural-language explanations. However, LLM-only approaches do not explicitly model accounting constraints, lack structural awareness of relationships among

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entries and accounts, and may produce explanations that are not faithful to the underlying transaction structure.

Graph neural networks (GNNs) have emerged as powerful tools for fraud detection in financial networks [5]. By representing entities and transactions as nodes and edges in a graph, GNNs can capture complex relational patterns invisible to methods operating on individual records [6].

In this paper, we propose the Constraint-Guided LLM-GNN (CG-LGN) framework, which combines three modules to address these limitations. Our contributions are:

- We construct a heterogeneous graph representation of journal entry data and apply a relational GCN (R-GCN) [7] for structural anomaly scoring.
- We design an accounting constraint module that encodes five domain-specific rules as penalty terms, providing interpretable regularization of anomaly scores.
- We integrate an LLM-based explanation generator that converts detected anomalies into auditor-readable reports as a post-hoc interpretability layer.
- We demonstrate on synthetic data with six anomaly types that the GNN and constraint modules jointly outperform individual baselines in PR-AUC, Precision@K, and false positive control.

II. Related Work

Journal Entry Testing. Huang et al. [8] integrated NLP and deep learning for manual journal entry evaluation. Bakumenko et al. [9] explored LLM embeddings for encoding non-semantic financial data features. Mashiko et al. [10] proposed a federated learning system for multi-organization anomaly detection on journal entry data without sharing raw records.

GNNs for Anomaly Detection. Kipf and Welling [11] introduced Graph Convolutional Networks for semi-supervised node classification. Wang et al. [12] proposed the Heterogeneous Attention Network using hierarchical attention over meta-paths.

LLMs for Anomaly Detection. Gu et al. [13] discussed foundation models as AI co-pilots in auditing. Tsai et al. [14] proposed AnoLLM, which serializes tabular data into text for LLM-based unsupervised anomaly detection. Recent work also broadens the methodological basis for calibrated risk ranking under multiple objectives. Causal generative re-ranking and risk-adaptive pricing formulate ranking policies that balance utility, exposure, and risk under changing operating conditions [15,16]. Their emphasis on context-sensitive reordering and explicit risk trade-offs complements audit scoring, where a review queue must distinguish high-priority entries while limiting false positives.

Related AI applications highlight how uncertainty-aware multimodal analysis can surface latent risk signals from heterogeneous evidence [17,18]. Generate-and-verify policy design, decentralized LLM inference, parameter-efficient adaptation, agent redundancy elimination, and SLO-driven autoscaling further show how LLM-enabled systems can retain verifiability and responsiveness under distributed resource constraints [19–23].

Reliable anomaly detection must also tolerate rare events, evidence gaps, and shifting operational context. Target-aware augmentation for rare-event prediction, evidence-verified root cause analysis, confidence-conflict routing, and structure-grounded document reasoning each emphasize that a system should condition decisions on observed support and uncertainty rather than on an unconstrained score alone [24–27]. Drift-aware mitigation in heterogeneous split federated learning further illustrates the value of adapting collaborative models to nonstationary conditions [28].

Deployment-oriented studies extend these concerns to persistent agent state, topology-aware localization, and efficient model serving [29–32]. Work on cost-aware cross-cloud federated learning, cold-start mitigation, semantic schema-drift repair, risk-calibrated edge-cloud scheduling, retrieval-augmented log anomaly detection, and learning-augmented serverless inference likewise treats system constraints and operational drift as first-class inputs to decision-making [33–38]. Together, these directions motivate the CG-LGN design: structural scores are fused with explicit accounting constraints so that the resulting audit priorities remain both adaptive and traceable.

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III. Methodology

A. Overview

Figure 1 illustrates the CG-LGN framework. Given a set of journal entries, the system proceeds through four stages: (1) heterogeneous graph construction, (2) GNN-based structural anomaly scoring, (3) accounting constraint evaluation, and (4) risk score fusion with LLM-generated explanations.

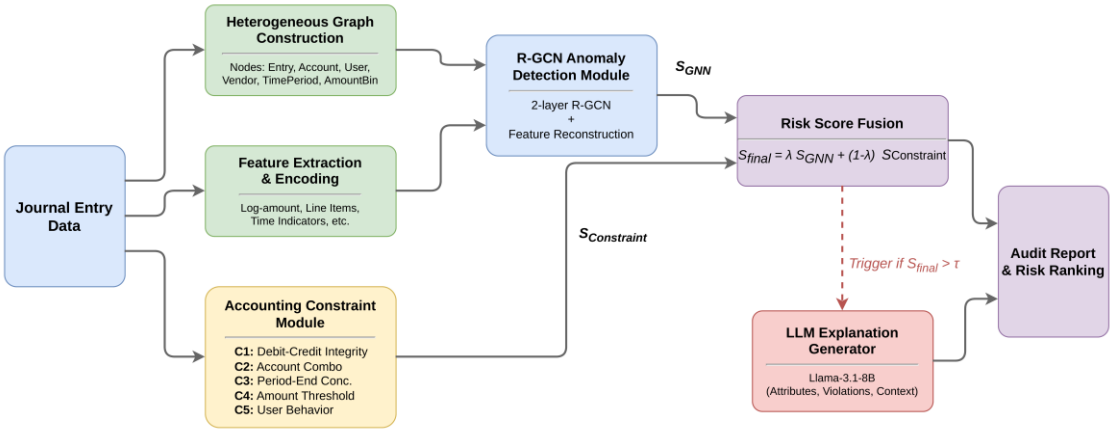


Figure 1. Architecture of the proposed CG-LGN framework. Journal entry data is transformed into a heterogeneous graph and encoded into feature vectors. The R-GCN module produces s_{GNN}; the accounting constraint module evaluates five rules (C1–C5) to produce $s_{\text{Constraint}}$. The fused score $s_{\text{final}} = \lambda\, s_{\text{GNN}} + (1-\lambda)\, s_{\text{Constraint}}$ triggers the LLM explanation generator for entries exceeding threshold τ.

B. Heterogeneous Graph Construction

We model journal entry data as a heterogeneous graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{T}_V, \mathcal{T}_E)$ with six node types $\mathcal{T}_V = \{\text{Entry, Account, User, Vendor, TimePeriod, AmountBin}\}$ and five edge types $\mathcal{T}_E$: entry-uses-account, entry-created-by-user, entry-in-period, entry-involves-vendor, and debit-credit-pairing between accounts.

Each entry node v_i is initialized with a feature vector $\mathbf{x}_i \in \mathbb{R}^d$ encoding the log-transformed entry amount, number of line items, posting hour, day-of-week indicators, and binary flags for manual entry and period-end posting. Other entity nodes are initialized with learnable embeddings.

C. GNN Anomaly Detection Module

We employ R-GCN to perform message passing on the heterogeneous graph. For each relation type $r \in \mathcal{T}_E$, the update rule at layer l is:

$$\mathbf{h}_i^{(l+1)} = \sigma\left(\sum_{r\in\mathcal{T}_E}\sum_{j\in\mathcal{N}_i^r}\frac{1}{|\mathcal{N}_i^r|}\mathbf{W}_r^{(l)}\mathbf{h}_j^{(l)} + \mathbf{W}_0^{(l)}\mathbf{h}_i^{(l)}\right) \tag{1}$$

where $\mathcal{N}_i^r$ denotes the neighbors of node i under relation r, $\mathbf{W}_r^{(l)}$ and $\mathbf{W}_0^{(l)}$ are learnable weight matrices, and σ is the ReLU activation. We stack two R-GCN layers to obtain entry-node embeddings $\mathbf{z}_i = \mathbf{h}_i^{(2)}$.

The GNN anomaly score is computed via reconstruction error:

$$s_{\text{GNN}}(v_i) = \| \mathbf{x}_i - g_\phi(\mathbf{z}_i) \|_2^2 \tag{2}$$

where g_ϕ is a learned decoder. The R-GCN is trained transductively on the full unlabeled graph; the training data may contain a small fraction ($\leq 2\%$) of contaminated anomalous entries. No anomaly labels are used for training; labels are only used for tuning α_k, λ, and τ on the validation set (Section III-E).

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D. Accounting Constraint Module

We define five constraint functions that encode domain-specific accounting rules. Each produces a penalty score $c_k(v_i) \in [0,1]$ for entry v_i.

C1: Debit-Credit Integrity. For entry v_i with debit total D_i and credit total C_i:

$$c_1(v_i) = \min\left(\frac{|D_i - C_i|}{\max(D_i, C_i) + \epsilon},\ 1\right) \tag{3}$$

This constraint captures bookkeeping integrity violations; fraud-related entries typically remain balanced, so C1 serves as a data-integrity check.

C2: Unusual Account Combination. We precompute the empirical frequency $f(a_d, a_c)$ of each debit-credit account pair from historical data:

$$c_2(v_i) = 1 - \frac{\log(f(a_d, a_c) + 1)}{\log(f_{\max} + 1)} \tag{4}$$

C3: Period-End Concentration. Let $r_p = n_w/N_p$ denote the fraction of entries posted in the last three business days of a reporting period, and r_{base} the historical baseline ratio:

$$c_3(v_i) = \begin{cases} \max\left(0,\ \min\left(\frac{r_p - r_{\text{base}}}{\gamma\, r_{\text{base}} + \epsilon},\ 1\right)\right) & \text{if last 3 days} \\ 0 & \text{otherwise} \end{cases} \tag{5}$$

where $\gamma = 1$. This provides a contextual signal useful when combined with other constraints and graph-structure information.

C4: Amount Threshold Proximity. Let T denote a set of known approval thresholds ($10K, $50K, $100K). For entry amount m_i:

$$c_4(v_i) = \max_{t \in T}\left(1 - \frac{|m_i - t|}{\delta \cdot t}\right)^+ \tag{6}$$

where $\delta = 0.05$ defines a 5% proximity band and $(\cdot)^+ = \max(\cdot, 0)$.

C5: User Behavior. Let $\text{role}(u_i) \in \{0,1\}$ indicate whether the creator is a non-financial user, $\text{offhours}(v_i) \in \{0,1\}$ whether the entry was posted outside business hours, and q_{95} the 95th percentile of entry amounts:

$$c_5(v_i) = \text{role}(u_i) \cdot \text{offhours}(v_i) \cdot \mathbf{1}[m_i > q_{95}] \tag{7}$$

The aggregate constraint score is:

$$s_{\text{Constraint}}(v_i) = \sum_{k=1}^{5} \alpha_k \cdot c_k(v_i) \tag{8}$$

where α_k are weights tuned via grid search on a held-out validation set containing 50 labeled anomalies. The individual constraint functions c_k are fully rule-based; only α_k selection uses labels.

E. Risk Score Fusion

The final anomaly score integrates both modules:

$$s_{\text{final}}(v_i) = \lambda \cdot \hat{s}_{\text{GNN}}(v_i) + (1 - \lambda) \cdot \hat{s}_{\text{Constraint}}(v_i) \tag{9}$$

where $\hat{s}$ denotes min-max normalized scores. In our experiments, $\lambda = 0.6$ yields the best validation PR-AUC. A sensitivity analysis over $\lambda \in \{0.3, 0.4, 0.5, 0.6, 0.7, 0.8\}$ shows that PR-AUC remains within ± 0.02 of the optimum for $\lambda \in [0.5, 0.7]$.

F. LLM Explanation Generator

For entries with $s_{\text{final}} > \tau$, we invoke Llama-3.1-8B with a structured prompt containing the entry attributes, top contributing constraint violations, and neighboring entries sharing accounts or users. The LLM generates a natural-language explanation and suggests relevant audit procedures. This module does not influence s_{final}; it serves exclusively as a post-hoc interpretability layer.

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IV. Experiments

A. Dataset

We construct a synthetic journal entry dataset emulating key characteristics of double-entry bookkeeping. The dataset contains 50,000 entries across 120 accounts, 45 users (35 financial, 10 non-financial), 12 monthly periods, and 200 vendors. Entry amounts follow a log-normal distribution with $\mu = 8.5,\ \sigma = 1.8$ (median $\approx$\$4,900).

We inject six anomaly types at a 2% contamination rate (1,000 anomalous entries): (1) period-end large adjustments (200), (2) split transactions below approval thresholds (180), (3) unusual account combinations (170), (4) fictitious vendor payments (160), (5) premature revenue recognition (150), and (6) improper expense capitalization (140). We designate 50 labeled anomalies and 500 labeled normal entries as a validation set for tuning α_k, λ, and τ; these are excluded from final evaluation. Final metrics are computed on the remaining 49,450 test entries.

B. Baselines and Setup

We compare CG-LGN against six baselines: (1) JET Rules: traditional rule-based filtering; (2) Isolation Forest (unsupervised); (3) One-Class SVM (unsupervised); (4) XGBoost trained on the same labeled validation subset, with hyperparameters selected by cross-validation; (5) GNN-only: our R-GCN module without constraints; (6) LLM-only: prompt-based detection where the model outputs a risk score in $[0,1]$ for each entry.

The R-GCN uses 2 layers with hidden dimension 64, trained for 200 epochs with Adam (lr = 10^{-3}). All experiments are repeated 5 times with different random seeds; we report mean values.

C. Main Results

Table 1 presents the main results. CG-LGN achieves the highest PR-AUC (0.49) and AUC (0.96), improving PR-AUC from 0.29 (GNN-only) to 0.49, a 0.20 absolute gain (69% relative improvement). CG-LGN reduces FP/1K to 8, which is 47% lower than the GNN-only baseline (15) and 94% lower than JET Rules (130).

Table 1. Main results on the synthetic journal entry dataset. P@100 denotes Precision@100. FP/1K denotes false positives per 1,000 entries at the validation-selected threshold τ. ↓ indicates lower is better.

Method	AUC	PR-AUC	P@100	FP/1K↓
JET Rules	0.59	0.03	0.04	130
Isolation Forest	0.76	0.07	0.19	51
One-Class SVM	0.77	0.07	0.22	30
XGBoost	0.88	0.19	0.51	20
GNN-only	0.91	0.29	0.73	15
LLM-only	0.89	0.24	0.60	15
CG-LGN (Ours)	**0.96**	**0.49**	**0.93**	**8**

Figure 2 shows the Precision-Recall curves and Precision@K. CG-LGN maintains higher precision at all recall levels. At $K = 100$, CG-LGN achieves P@100 $= 0.93$, meaning 93 of 100 reviewed entries are true anomalies, compared to 0.73 for GNN-only and 0.04 for JET Rules.

Figure 3 provides deeper analysis. The heatmap (left) shows per-anomaly-type recall at each method's validation-selected threshold τ, revealing that CG-LGN consistently outperforms baselines across all six anomaly types. The scatter plot (right) visualizes PR-AUC vs. FP/1K, with the FP/1K axis inverted so that better methods appear in the upper-right region.

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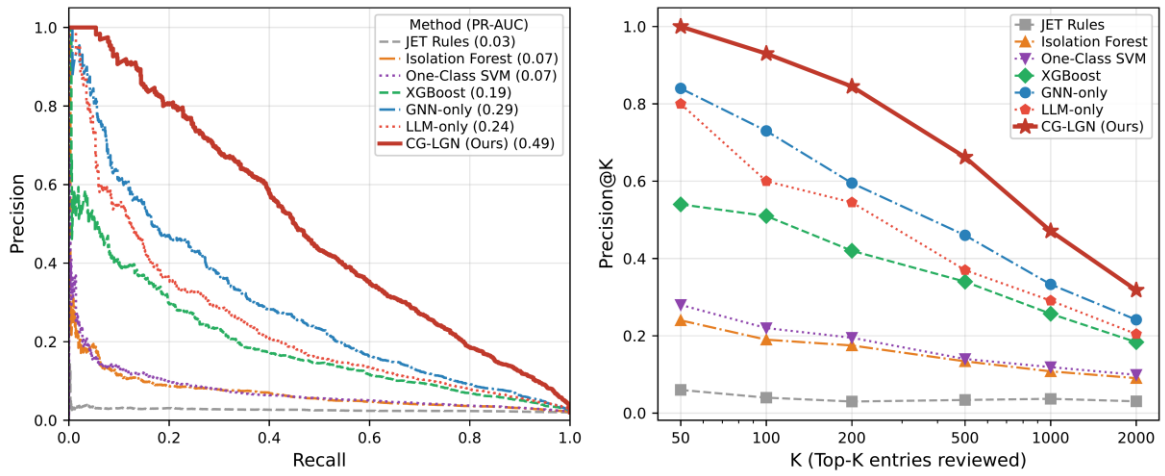


Figure 2. Left: Precision-Recall curves for all methods. Right: Precision@K as a function of the number of top-ranked entries reviewed. Precision@K generally decreases as K grows because lower-ranked entries are gradually included in the review set.

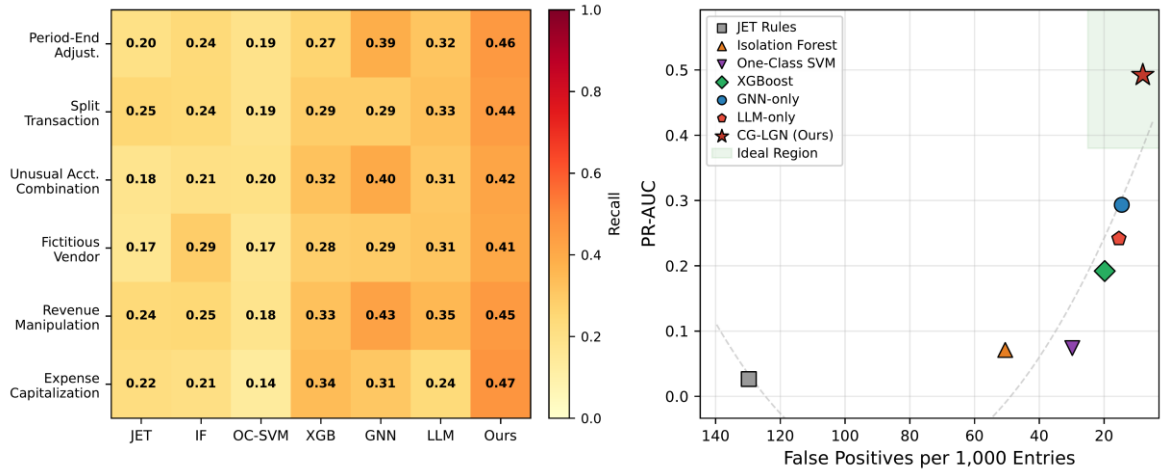


Figure 3. Left: Per-anomaly-type recall heatmap at each method's validation-selected threshold τ. Right: PR-AUC vs. FP/1K (axis inverted; upper-right is ideal). CG-LGN achieves the best trade-off.

D. Ablation Study

Table 2 presents ablation results. Removing the GNN module reduces PR-AUC by 43%, confirming the importance of structural modeling. Removing the constraint module decreases PR-AUC by 14% and increases FP/1K by 50%, validating its role in false positive suppression. The LLM module does not affect detection metrics; its contribution is assessed in Section IV-E.

Table 2. Ablation study. The LLM module does not affect detection scores; its quality is evaluated separately in Table 3.

Configuration	PR-AUC	P@100	FP/1K↓
Full CG-LGN	**0.49**	**0.93**	**8**
w/o GNN	0.28	0.55	24
w/o Constraint	0.42	0.85	12
w/o LLM Explanation	0.49	0.93	8

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Table 3. Explanation quality ratings (1–5 Likert, higher is better) by two domain experts on 50 true positive detections. Inter-rater agreement (ICC) was 0.72.

Condition	Relevance	Accuracy	Actionability
CG-LGN Grounded	$3.9_{\pm 0.7}$	$3.7_{\pm 0.8}$	$3.6_{\pm 0.9}$
LLM-only	$3.4_{\pm 0.9}$	$3.0_{\pm 1.0}$	$2.9_{\pm 1.0}$
Constraint-only	$3.6_{\pm 0.6}$	$3.5_{\pm 0.7}$	$2.4_{\pm 0.8}$

E. Explanation Quality

We sampled 50 true positive detections covering all six anomaly types and asked two auditing domain experts to rate each explanation on a 1–5 Likert scale. The explanations were anonymized and presented in randomized order. We compare three conditions: (a) CG-LGN grounded explanations (with constraint violations and graph context), (b) LLM-only explanations (entry attributes only), and (c) constraint-only explanations (rule violation list). Results are shown in Table 3.

CG-LGN grounded explanations outperform both alternatives, particularly in actionability. Experts noted that constraint-based grounding improved accuracy over LLM-only outputs, which sometimes produced plausible but unfaithful reasoning. Constraint-only explanations scored well on accuracy but lacked actionable audit procedure suggestions.

V. Discussion and Conclusions

The experimental results demonstrate that combining GNN-based structural analysis with explicit accounting constraints yields meaningful improvements over either approach alone. The constraint module effectively reduces false positives by encoding domain knowledge that is difficult for data-driven models to learn from limited examples. Because some injected anomaly types are aligned with common audit rules, the constraint module may benefit from this correspondence; future work should evaluate generalization to unseen anomaly types.

We acknowledge several limitations. First, our evaluation uses synthetic data, which may not capture the full complexity of real-world accounting systems; validation on real enterprise data remains important. Second, the graph model is evaluated in a transductive setting; future work should examine inductive generalization to new entries and entities. Third, the LLM module operates as a post-hoc addition; tighter integration could improve both detection and explanation quality. Finally, the constraint weights α_k and fusion parameter λ are tuned on a small validation set, and their optimal values may vary across enterprises.

In summary, CG-LGN integrates graph neural networks, accounting domain constraints, and LLM-based explanations for journal entry anomaly detection. The GNN and constraint modules jointly improve anomaly ranking and false-positive control, while the LLM module enhances interpretability by generating auditor-readable explanations grounded in structural and constraint-based evidence. Future work could explore temporal graph models for evolving transaction patterns, federated learning for multi-client settings, and tighter coupling between the LLM and the scoring pipeline.

References

1. K. C. Droste and J. Tritschler, *Journal entry testing*, 2nd ed. IDW Verlag, Düsseldorf, 2024.
2. M. Schreyer, T. Sattarov, D. Borth, A. Dengel and B. Reimer, "Detection of Anomalies in Large Scale Accounting Data Using Deep Autoencoder Networks", arXiv preprint arXiv:1709.05254, 2017.
3. F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in *IEEE international conference on data mining (ICDM)*, IEEE, 2008, pp. 413–422.

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4. G. Singh, P. Singh and M. Singh, "Advanced Real-Time Fraud Detection Using RAG-Based LLMs", arXiv preprint arXiv:2501.15290, 2025.
5. D. Cheng, Y. Zou, S. Xiang, and C. Jiang, "Graph neural networks for financial fraud detection: A review," *Frontiers of Computer Science*, vol. 19, no. 9, p. 199609, 2025.
6. H. Qiao, H. Tong, B. An, I. King, C. Aggarwal, and G. Pang, "Deep graph anomaly detection: A survey and new perspectives," *IEEE Transactions on Knowledge and Data Engineering*, vol. 37, no. 9, pp. 5106–5126, 2025.
7. M. Schlichtkrull, T. N. Kipf, P. Bloem, R. van den Berg, I. Titov, and M. Welling, "Modeling relational data with graph convolutional networks," in *European semantic web conference (ESWC)*, Springer, 2018, pp. 593–607.
8. Q. Huang, H. K. Duan, and M. A. Vasarhelyi, "Manual journal entry testing: Integrating natural language processing and deep learning," *Intelligent Systems in Accounting, Finance and Management*, vol. 32, no. 3, p. e70016, 2025.
9. A. Bakumenko, K. Hlaváčková-Schindler, C. Plant, and N. C. Hubig, "Advancing anomaly detection: Non-semantic financial data encoding with LLMs," *IEEE Access*, vol. 13, pp. 146757–146771, 2025.
10. S. Mashiko, Y. Kawamata, T. Nakayama, et al., "Anomaly detection in double-entry bookkeeping data by federated learning system with non-model sharing approach," *Scientific Reports*, vol. 15, p. 42208, 2025.
11. T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *International conference on learning representations (ICLR)*, 2017.
12. X. Wang et al., "Heterogeneous graph attention network," in *The world wide web conference (WWW)*, ACM, 2019, pp. 2022–2032.
13. H. Gu, M. Schreyer, K. Moffitt, and M. A. Vasarhelyi, "Artificial intelligence co-piloted auditing," *International Journal of Accounting Information Systems*, vol. 54, p. 100698, 2024.
14. C.-P. Tsai, G. Teng, P. Wallis, and W. Ding, "AnoLLM: Large language models for tabular anomaly detection," in *International conference on learning representations (ICLR)*, 2025.
15. M. Momma, A. B. Garakani, N. Ma, and Y. Sun, "Multi-Objective Ranking via Constrained Optimization," in Companion Proc. Web Conf. 2020 (WWW '20 Companion), Taipei, Taiwan, 2020, pp. 111–112, doi: 10.1145/3366424.3382723.
16. Q. Wu, D. Gao, Q. Lin, Z. Wu, and M. Z. Shou, "GUI-Narrator: Detecting and Captioning Computer GUI Actions," in Proc. 33rd ACM Int. Conf. Multimedia (MM '25), Dublin, Ireland, 2025, pp. 3683–3692, doi: 10.1145/3746027.3755150.
17. L. Deng, P. Rattadilok, and R. Xiong, "A Machine Learning-Based Monitoring System for Attention and Stress Detection for Children with Autism Spectrum Disorders," in Proc. Int. Conf. Intelligent Medicine and Health, 2022, doi: 10.1145/3484377.3484381.
18. K. Kaffes, N. Yadwadkar, and C. Kozyrakis, "Centralized Core-Granular Scheduling for Serverless Functions," in Proc. ACM Symp. Cloud Computing (SoCC), 2019, doi: 10.1145/3357223.3362709.
19. Y. Wu, T. T. A. Dinh, G. Hu, M. Zhang, Y. M. Chee, and B. C. Ooi, "Serverless Data Science—Are We There Yet? A Case Study of Model Serving," in Proc. 2022 Int. Conf. Management of Data (SIGMOD/PODS '22), Philadelphia, PA, USA, 2022, pp. 1866–1875, doi: 10.1145/3514221.3517905.
20. Z. Wang, S. Shu, Y. Wang, I. H. Lai, and C. C. Peng, "Communication-Efficient Decentralized LLM Inference over Low-Bandwidth Distributed Nodes," 2026, doi: 10.13140/RG.2.2.34464.96000.
21. C. He, Y. Gong, Y. Ma, B. Zhang, and S. Wang, "Subspace-Deconvolution Parameterization for Efficient Large Language Model Adaptation," 2026, doi: 10.13140/RG.2.2.34546.88008.
22. Z. Xiao, Q. Guo, Y. Wang, B. Wang, and H. Lu, "AgentDropout: Dynamic Redundancy Elimination for Multi-Agent Collaboration Efficiency," 2026, doi: 10.13140/RG.2.2.15489.01125.
23. S. Wang, X. Sun, S. Y. Huang, B. Su, S. Xu, and N. Awate, "SLO-Scaler: Uncertainty-Aware SLO-Driven Autoscaling for Microservices," arXiv preprint arXiv:2608.18390, 2026.
24. D. Huang, N. Zhao, W. Liu, and N. Sang, "Target-Aware Augmentation for Rare-Event Prediction under Tabular Covariate Shift," 2026, doi: 10.13140/RG.2.2.32056.51202.
25. W. Ma, K. Yang, Z. Liu, R. Jiang, and Y. Jiang, "VerifyRCA: LLM-Guided but Evidence-Verified Root Cause Analysis for Microservice Backends," in Proc. 11th Int. Conf. Electronic Technology and Information Science (ICETIS), Guiyang, China, 2026, pp. 322-327, doi: 10.1109/ICETIS70504.2026.11633469.

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26. R. Fang et al., "Confidence-Conflict Routing Between Parametric Memory, RAG, and Long-Context LLMs," in Proc. 11th Int. Conf. Electronic Technology and Information Science (ICETIS), Guiyang, China, 2026, pp. 328–333, doi: 10.1109/ICETIS70504.2026.11633419.
27. J. Zhou, Z. Zhu, A. Zhu, S. Wang, and Y. Zheng, "Structure-Grounded Document Agents for Faithful Long-Context Reasoning," 2026, doi: 10.13140/RG.2.2.25574.38723.
28. J. Zhang, B. Chen, Y. Tang, F. Chen, Y. Zhan, and Z. Hu, "Drift-Aware Straggler Mitigation for Heterogeneous Split Federated Learning via Proactive Cut-Layer and Asynchronous Server Updates," 2026, doi: 10.13140/RG.2.2.26441.10080.
29. B. Su, S. Wang, S. Pi, Y. Liu, and Z. Jiang, "Memcurrency: Memory Decay and Refresh Mechanism for Long-Running LLM Agents," in Proc. 11th Int. Conf. Electronic Technology and Information Science (ICETIS), 2026, pp. 316-321.
30. W. Ma, Z. Liu, R. Jiang, and Y. Jiang, "TopoRCA-Lite: Dynamic Topology-Aware Root Cause Localization under Limited Observability," in Proc. 6th Int. Conf. Machine Learning and Intelligent Systems Engineering (MLISE), 2026, pp. 676-681.
31. T. Xia, X. Huang, Y. Zhou, F. Chang, and L. Ren, "Coordinating Training and Inference in Heterogeneous Cloud GPU Clusters with Learned Interference Models," in Proc. 8th Int. Conf. Internet of Things, Automation and Artificial Intelligence (IoTAAI), 2026, pp. 104-110.
32. J. Sun, Z. Hu, J. Zhang, Z. Yang, and C. Zhang, "CacheCast: Learning-Guided KV Cache Placement for Multi-Tenant Cloud LLM Serving," in Proc. 8th Int. Conf. Internet of Things, Automation and Artificial Intelligence (IoTAAI), 2026, pp. 647-653.
33. J. Jiang, J. Hu, and Y. Lyu, "Cost-Aware Cross-Cloud Federated Learning with Learned Client Routing and Aggregation Selection," in Proc. 8th Int. Conf. Internet of Things, Automation and Artificial Intelligence (IoTAAI), 2026, pp. 111-116.
34. S. Zhang, S. Shawulieti, and S. Pi, "Cold-Start Budgeting for Serverless ML Inference via Workload Forecasting and Model State Tiering," in Proc. 3rd Int. Conf. Image Processing and Artificial Intelligence (ICIPAI), 2026, pp. 345-351.
35. L. Tang, Y. Mei, X. Ren, Y. Ke, Z. Ke, and Z. Bian, "DriftMender: LLM-Guided Semantic Repair for Schema Drift in Real-Time ETL Pipelines," in Proc. 6th Int. Conf. Artificial Intelligence, Big Data and Algorithms (CAIBDA), 2026, pp. 1528-1533.
36. X. Huang, T. Xia, Y. Zhou, J. Liao, L. Ren, and F. Chang, "Risk-Calibrated Uncertainty-Aware Edge-Cloud Scheduling for Reliable Multi-Tenant ML Inference," in Proc. 6th Int. Conf. Electronics, Circuits and Information Engineering (ECIE), Suzhou, China, 2026, pp. 573-578, doi: 10.1109/ECIE69549.2026.11565531.
37. Y. Yang, Z. Liu, C. Chen, S. Dang, and K. Wu, "Drift-Aware Retrieval-Augmented Log Anomaly Detection for Cloud AIOps," in Proc. 11th Int. Conf. Electronic Technology and Information Science (ICETIS), Guiyang, China, 2026, pp. 205-210, doi: 10.1109/ICETIS70504.2026.11633481.
38. J. Jiang, J. Hu, and Y. Lyu, "Learning Augmented Cold Start Mitigation for Serverless Machine Learning Inference," in Proc. 7th Int. Conf. Big Data and Artificial Intelligence and Software Engineering (ICBASE), Shenyang, China, 2026, pp. 295-300, doi: 10.1109/ICBASE70763.2026.11619480.

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